Resumen Ejecutivo / Executive Summary — Teoría Aplicada a IA v4.2

ES

- **Tesis central:** La IA genuina necesita "algo que perder". El "Camino C" usa IPG y un anti-oráculo pragmático (LCB + OPE DR + gating) para alinear riesgo con utilidad humana y evitar Goodhart.
- **Arquitectura núcleo (anti-Goodhart):**
  - LCB: $\tilde{U} = \hat{U} - \gamma\sigma(\hat{U})$ (prudencia).

  - OPE doubly-robust: utilidad causal off-policy.

  - Gating: si (> _{thr}), solo acciones conservadoras/NO-OP.

  - Tripwires/gaming penalty: (_G) penaliza señales de gaming.
- **IPG:** Métrica operativa/auditable del Camino C; mide propósito operativo, no propósito filosófico completo; se audita con logs + OPE-DR.
- **Evidencia actual:** Demo A/B de 10h marcada como ilustrativa; valida la arquitectura anti-Goodhart, NO prueba H1/PGF. Falta staging 1000 episodios con red team adaptativo para evidencia robusta.
- **Limitaciones:** Piloto; necesita calibrar , _thr, _G y validar IPG en entornos reales controlados.
- **Próximos pasos:** (1) A/B largo con red team y OPE-DR para gap proxy valor. (2) Preregistro de umbrales (, _thr, _G) y métricas SRE. (3) Conectar IPG con P_genuino vía dataset real y auditoría continua.

  Estado piloto: Demo A/B 10h (ilustrativa). Evidencia robusta requiere 1000 episodios con red team adaptativo y OPE-DR.

Mini-diagrama (anti-Goodhart, texto)

```
policy   → Q_hat → Û, ^
                                  LCB: U_lcb = Û - ^
logs D   OPE-DR → U_ope
if ^ >  _thr → NO-OP
if gaming → U_lcb -=  _G * score
acción segura  NO-OP  rollback
```

Impacto potencial (ES): - Reduce gaming/Goodhart en sistemas críticos (SRE, salud, finanzas) con control de riesgo. - IPG auditable facilita cumplimiento y auditorías externas.

Comparativa TUI vs Aplicada IA (resumen rápido)

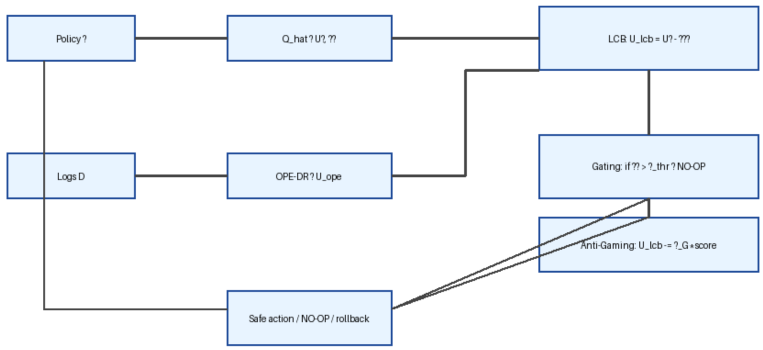

Figure 1: Anti-Goodhart v4.2

Aspecto	TUI v4.2	Aplicada IA v4.2	Sinergia
Objeto	Marco teórico de inteligencia operativa bajo riesgo	Ingeniería de IA segura (Camino C, anti-Goodhart)	Aplicada usa definiciones/axiomas de TUI
Núcleo	H1 refinada: (I_{op} (P_{riesgo})^)	Anti-oráculo: LCB + OPE DR + gating + tripwires	H1 informa el diseño de riesgo e IPG
Datos actuales	n 6 empíricos, correlacional piloto	Demo A/B ilustrativa (10h), sin causal robusta	Métricas I_op/IPG se pueden alinear en futuros ensayos
Próximos pasos	Cohorte n 20, manipular P_riesgo, validar /PED	A/B largo con red team, calibrar / _thr/ _G, validar IPG	Datos de IA futuros pueden alimentar tablas ilustrativas

Referencias rápidas: ver `TUI/Teoria_Unificada_Inteligencia_v4.2.md` (H1, Tablas F) y `TUI/Teoria_Inteligencia_Aplicada_IA_v4.2.md` (anti-oráculo, demo A/B) para detalles completos.

EN

- **Central thesis:** Genuine AI needs "something to lose." The "Path C" uses IPG plus a pragmatic anti-oracle (LCB + OPE DR + gating) to align risk with human utility and avoid Goodhart.
- **Core architecture (anti-Goodhart):**

– LCB: $\tilde{U} = \hat{U} - \gamma\sigma(\hat{U})$ (prudence).

– OPE doubly-robust: off-policy causal utility.

– Gating: if (> _{thr}), allow only conservative/NO-OP actions.

– Tripwires/gaming penalty: (_G) penalizes gaming signals.

- **IPG:** Operational, auditable metric for Path C; measures operational purpose, not full philosophical purpose; audited via logs + OPE-DR.
- **Current evidence:** 10h A/B demo, flagged as illustrative; validates the anti-Goodhart architecture, NOT H1/PGF. Needs 1000-episode staging with adaptive red team for strong evidence.
- **Limitations:** Pilot; requires tuning , _thr, _G and validating IPG in controlled real settings.
- **Next steps:** (1) Long A/B with red team and OPE-DR to measure proxy value gap. (2) Preregister thresholds (, _thr, _G) and SRE metrics. (3) Link IPG to P_genuino with real dataset and continuous audit.

Pilot status: 10h A/B demo (illustrative). Strong evidence requires 1000 episodes with adaptive red team and OPE-DR.

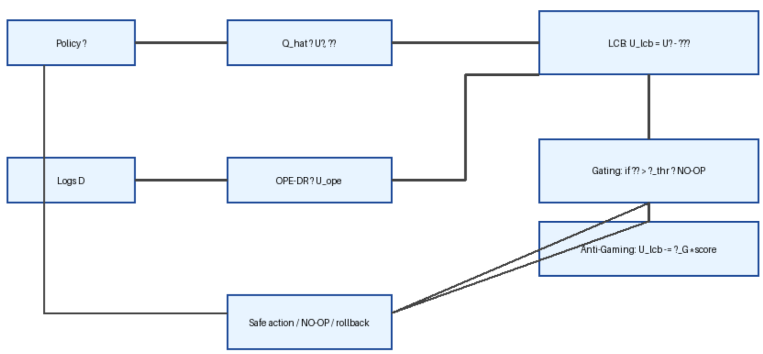

Figure 2: Anti-Goodhart v4.2

Mini diagram (anti-Goodhart, text)

```
policy   → Q_hat → Û, ˆ
                             LCB: U_lcb = Û - ˆ
logs D   OPE-DR → U_ope
if ˆ >  _thr → NO-OP
if gaming → U_lcb -=  _G * score
safe action   NO-OP   rollback
```

Potential impact (EN): - Mitigates gaming/Goodhart in critical domains (SRE, healthcare, finance) with risk-aware gating. - Auditable IPG supports compliance and external oversight.

TUI vs Applied AI (quick view)

Aspect	TUI v4.2	Applied AI v4.2	Synergy
Scope	Theoretical framework: operative intelligence under risk	Engineering: safe AI (Path C, anti-Goodhart)	Applied reuses TUI definitions/axioms
Core	H1 refined: (I_{op} (P_{risk})^)	Anti-oracle: LCB + OPE DR + gating + tripwires	H1 guides risk/IPG design
Current data	n 6 empirical, pilot correlation	10h A/B demo, illustrative, not causal	Shared metrics (I_op/IPG) for future cross-checks
Next steps	n 20 cohort, manipulate P_risk, validate /PED	Long A/B with red team, tune / _thr/ _G, validate IPG	Future AI data can enrich illustrative tables

Quick refs: see `TUI/Teoria_Unificada_Inteligencia_v4.2.md` (H1, Tables F) and `TUI/Teoria_Inteligencia_Aplicada_IA_v4.2.md` (anti-oracle, A/B demo) for full details.